



Kuiper Belt dust on Earth

Some of the dust in the atmosphere of the Earth traces its origins to the very birth of the Solar System. <https://digit.in/may19-67>



Hubble Kuiper Belt Survey

The Hubble Space Telescope given the green signal to conduct the largest ever survey of the Kuiper Belt. <https://digit.in/may19-68>

An artist's conception of how the distant Sun would appear from the surface of Sedna.

THE KUIPER CLIFF

At the edge of the solar system, beyond a distance of 47.8 AU from the Sun, the number of objects suddenly drops off, and science does not have an explanation.

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B

yond the orbit of Neptune, are the remnants of the accretion disk that gave birth to the solar

system. This is the material that coalesced into all the planets and moons of the Solar System. Think of it as a second asteroid belt, but 20 times larger, and with 200 times more mass. The asteroid belt exists because of Jupiter, which continuously disturbed the material in the belt, preventing them from coalescing into a planet. Similarly, the Kuiper Belt is believed to be a consequence of the formation of Neptune, with the ice giant preventing these objects from forming another planet. While the asteroids are made up of mostly rock and iron, the objects in the Kuiper Belt are composed of

rocks, as well as frozen chunks of methane, ammonia and water. There are millions of these small ice chunks, and hundreds of thousands that are bigger than 100 kilometers in diameter. There are many names for this region, including the third zone, the trans-Neptunian belt and the Edgeworth-Kuiper Belt. It is most commonly known as the Kuiper Belt. The objects in the belt are known as Kuiper Belt Objects (KBOs). One AU is the distance between the Sun and the Earth, and serves as a convenient measure for describing the distance of objects within the solar system. Objects beyond the orbit of Neptune, including bodies in the Kuiper Belt (30-50 AU), a second region with fewer objects known as the scattered disk (50 to 2,000 AU), and the Oort Cloud beyond (2,000 to 200,000 AU) are together referred to as Trans Neptunian Objects (TNOs). The given distances are from the Sun. While the

asteroid belt, the Kuiper Belt and the scattered disk are all roughly torus shaped, the Oort Cloud is spherical.

In the 1940s and the 1950s, two astronomers, Kenneth Edgeworth and Gerard Kuiper both published scientific literature speculating on objects beyond Pluto. The most well known KBO is the demoted planet, Pluto. Pluto was the first KBO to be discovered in 1930, which was because of its brightness and size. The second KBO would only be discovered in 1992, an object called Albion. Eris, discovered in 2005, is the largest known object in the scattered disk. Over 2,000 TNOs are catalogued by science, but this is a fraction of the total number of objects expected to exist in the region. At about 50 AU, the number of objects drops off suddenly, which simply does not match the predictions and theoretical models of planetary formation. There are simply far fewer objects observed in the region than